1 Capacitors and Inductors

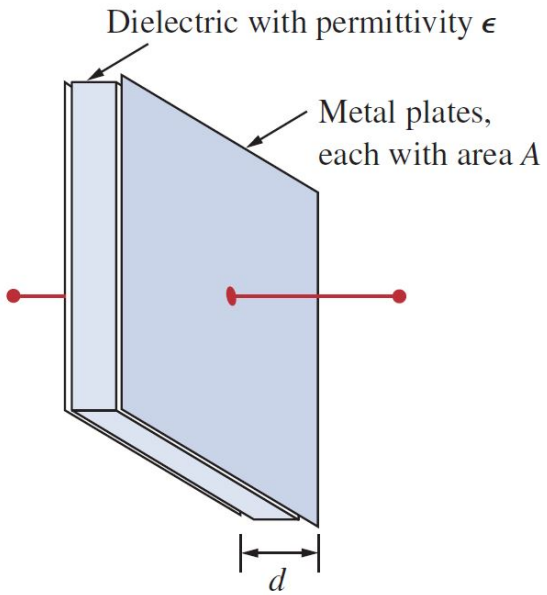
2 Capacitors
- Types of Capacitors
- Examples
- Series and Parallel Capacitors

Capacitors and Inductors

- Thus far, we have limited our study to resistive circuits.
- In this lecture, we introduce two new and important passive linear circuit elements: the capacitor and the inductor.
- Unlike resistors, which dissipate energy, capacitors and inductors do not dissipate but store energy, which can be retrieved at a later time.
- Therefore, capacitors and inductors are called storage elements.

Capacitors

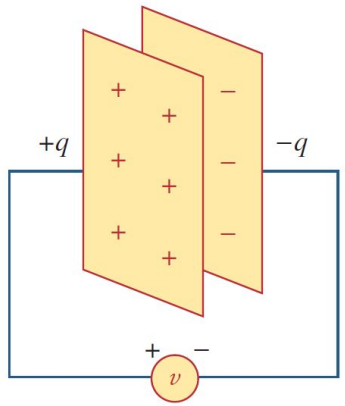
- A capacitor is a passive element designed to store energy in its electric field.
- Besides resistors, capacitors are the most common electrical components.
- Capacitors are used extensively in electronics, communications, computers, and power systems. For example, they are used in the tuning circuits of radio receivers and as dynamic memory elements in computer systems.
- A capacitor is typically constructed as illustrated below.



Capacitors

Capacitor

A capacitor consists of two conducting plates separated by an insulator (or dielectric).



- In many practical applications, the plates may be aluminum foil while the dielectric may be air, ceramic, paper, or mica.
- When a voltage source is connected to the capacitor, the source deposits a positive charge q on one plate and a negative charge $-q$ on the other. The capacitor is said to store the electric charge. The amount of charge stored, represented by q , is directly proportional to the applied voltage v.

Capacitors

- Therefore

$$q = Cv$$

where C, the constant of proportionality, is known as the capacitance of the capacitor. The unit of capacitance is the farad (F), in honor of the English physicist Michael Faraday (1791–1867).

Capacitance

Capacitance is the ratio of the charge on one plate of a capacitor to the voltage difference between the two plates, measured in farads (F).

Capacitors

- Although the capacitance C of a capacitor is the ratio of the charge q per plate to the applied voltage it does not depend on q or V. It depends on the physical dimensions of the capacitor.
- For instance, for the parallel-plate capacitor, the capacitance is given by

$$C = \frac{\epsilon A}{d}$$

where A is the surface area of each plate, d is the distance between the plates, and ϵ is the permittivity of the dielectric material between the plates.

Capacitors

- Although the equation $C = \frac{\epsilon A}{d}$ applies to only parallel-plate capacitors, we may infer from it that, in general, three factors determine the value of the capacitance:
  - The surface area of the plates – the larger the area, the greater the capacitance.
  - The spacing between the plates – the smaller the spacing, the greater the capacitance.
  - The permittivity of the material – the higher the permittivity, the greater the capacitance.

Capacitors

- Capacitors are commercially available in different values and types.
- Typically, capacitors have values in the picofarad (pF) to microfarad (μF) range. They are described by the dielectric material they are made of and by whether they are of fixed or variable type.
- The below figure shows the circuit symbols for fixed and variable capacitors. Note that according to the passive sign convention, $v > 0$ and $i > 0$ or if $i < 0$ and $v < 0$, the capacitor is being charged, and if $v \cdot i < 0$, the capacitor is discharging.

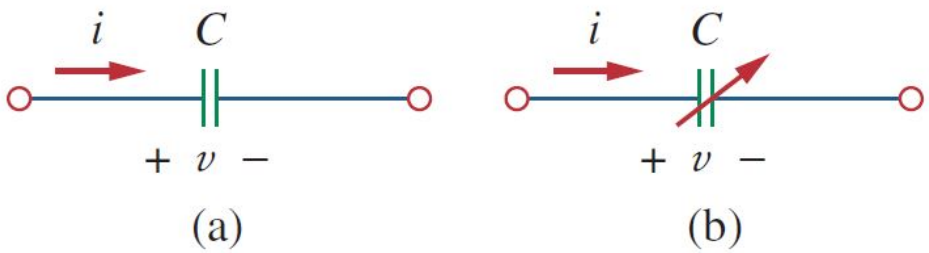


Figure: Circuit symbols for capacitors: (a) fixed capacitor, (b) variable capacitor

Capacitors

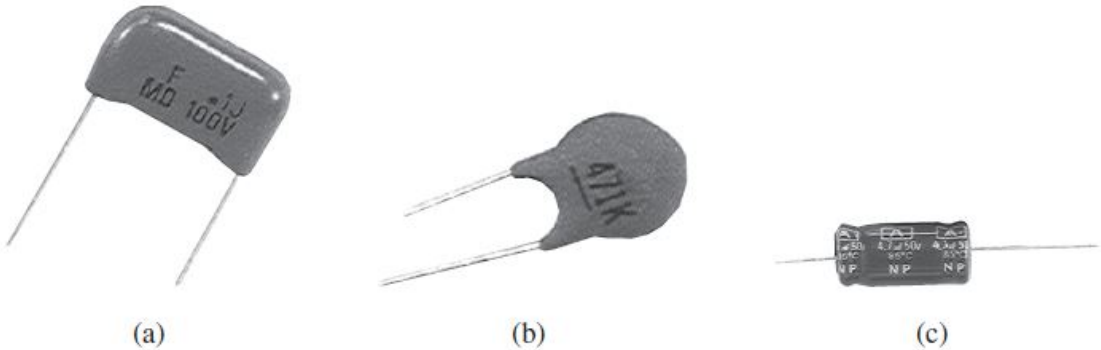


Figure: Fixed capacitors: (a) polyester capacitor, (b) ceramic capacitor, (c) electrolytic capacitor.

- The above figure shows common types of fixed-value capacitors. Polyester capacitors are light in weight, stable, and their change with temperature is predictable.
- Instead of polyester, other dielectric materials such as mica and polystyrene may be used. Film capacitors are rolled and housed in metal or plastic films. Electrolytic capacitors produce very high capacitance.

Capacitors

- The below figure shows the most common types of variable capacitors. The capacitance of a trimmer (or padder) capacitor is often placed in parallel with another capacitor so that the equivalent capacitance can be varied slightly.
- The capacitance of the variable air capacitor (meshed plates) is varied by turning the shaft. Variable capacitors are used in radio receivers allowing one to tune to various stations. In addition, capacitors are used to block DC, pass AC, shift phase, store energy, start motors, and suppress noise.

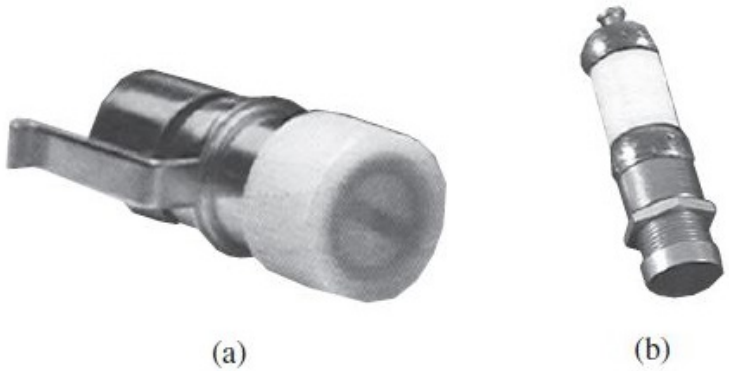


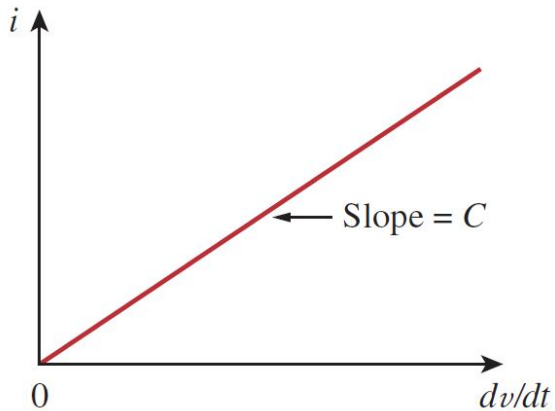
Figure: Variable capacitors: (a) trimmer capacitor, (b) filmtrim capacitor.

Capacitors

- In order to obtain the current-voltage relationship of the capacitor, we take the derivative of both sides of $q = Cv$. Since $i = dq/dt$, we have

$$i = C\frac{dv}{dt}$$

- This is the current-voltage relationship for a capacitor, assuming the passive sign convention. The relationship is illustrated in the figure below for a capacitor whose capacitance is independent of voltage.



Capacitors

- The voltage-current relation of the capacitor can be obtained by integrating both sides of $i = C\frac{dq}{dt}$. We have

$$v(t) = \frac{1}{C}\int_{-\infty}^{t} i(\tau)d\tau$$

or

$$v(t) = \frac{1}{C}\int_{t_0}^{t} i(\tau)d\tau + v(t_0)$$

where $v(t_0) = q(t_0)/C$ is the voltage across the capacitor at time t_0. Note from the above equation that the voltage of the capacitor depends on the past history of the capacitor current. Therefore, the capacitor has memory (unlike the resistor).

Capacitors

- The instantaneous power delivered to the capacitor is

$$p = vi = Cv\frac{dv}{dt}$$

Therefore the energy stored in the capacitor can be computed as

$$w = \int_{-\infty}^{t} p(\tau)d\tau = C\int_{-\infty}^{t} v\frac{dv}{d\tau}d\tau = C\int_{v(-\infty)}^{v(t)} vdv = \frac{1}{2}Cv^2\Big|_{v(-\infty)}^{v(t)}.$$

- We note that because the capacitor was uncharged at $t = -\infty$. Therefore

$$w = \frac{1}{2}Cv^2$$

- Note that since $q = Cv$, one can re-write the above equation as

$$w = \frac{q^2}{2C}$$

Capacitors

- These equations represent the energy stored in the electric field that exists between the plates of the capacitor. This energy can be retrieved, since an ideal capacitor cannot dissipate energy. In fact, the word capacitor is derived from this element's capacity to store energy in an electric field.
- The important properties of a capacitor can be listed as follows.
  1. When the voltage across a capacitor is not changing with time (i.e., DC voltage), the current through the capacitor is zero. Therefore **A capacitor is an open circuit to DC.** However, **if a battery (DC voltage) is connected across a capacitor, the capacitor charges.**
  2. The voltage on the capacitor must be continuous. **The voltage on a capacitor cannot change abruptly.**
  3. The ideal capacitor does not dissipate energy. It takes power from the circuit when storing energy in its field and returns previously stored energy when delivering power to the circuit.

Examples

Example

a) Calculate the charge stored on a 3pF capacitor with 20V across it.

b) Find the energy stored in the capacitor.

Examples

Example

a) Calculate the charge stored on a 3pF capacitor with 20V across it.

b) Find the energy stored in the capacitor.

Solution:

a) Since $q = Cv$, we have

$$q = 3 \times 10^{-12} \times 20 = 60pC$$

b) The energy stored is

$$w = \frac{1}{2}Cv^2 = \frac{1}{2} \times 3 \times 10^{-12} \times 400 = 600pJ$$

Examples

Example

The voltage across a $5\mu F$ capacitor is

$$v(t) = 10\cos 6000tV$$

Calculate the current through it.

Examples

Example

The voltage across a $5\mu F$ capacitor is

$$v(t) = 10\cos 6000t V$$

Calculate the current through it.

Solution: By definition, the current is

$$\begin{aligned} i(t) = C\frac{dv}{dt} &= 5 \times 10^{-6} \times \frac{d}{dt}(10\cos 6000t) \\ &= -5 \times 10^{-6} \times 6000 \times 10 \sin 6000t = -0.3 \sin 6000t A \end{aligned}$$

Examples

Example

Determine the voltage across a $2\mu F$ capacitor if the current through it is

$$i(t) = 6e^{-3000t} mA$$

Assume that the initial capacitor voltage is zero.

Examples

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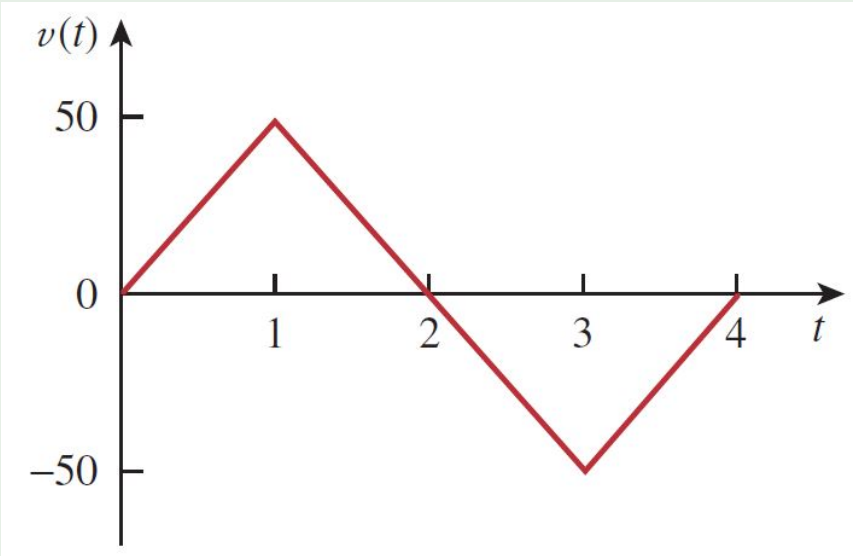
Solution: Since $v = \frac{1}{C}\int_0^t idt + v(0)$ and $v(0) = 0$,

$$\begin{aligned} v &= \frac{1}{2 \times 10^{-6}} \int_0^t 6e^{-3000t} dt \cdot 10^{-3} \\ &= \frac{3 \times 10^3}{-3000} e^{-3000t} \Big|_0^t = (1 - e^{-3000t})V \end{aligned}$$

Examples

Example

Determine the current through a $200\mu F$ capacitor whose voltage is shown in the figure below.



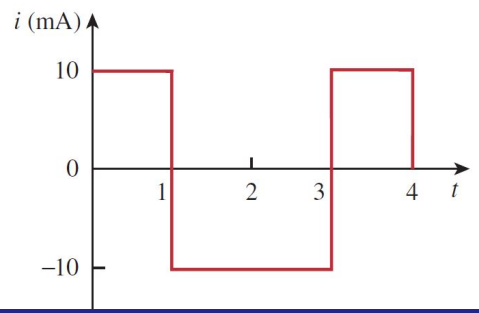
Examples

Solution: The voltage waveform can be described mathematically as

$$v(t) = \begin{cases} 50tV & 0 < t < 1 \\ 100 - 50tV & 1 \leq t < 3 \\ -200 + 50tV & 3 \leq t < 4 \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

Since $i = Cdv/dt$ and $C = 200\mu F$, we take the derivative of v with respect to t to get

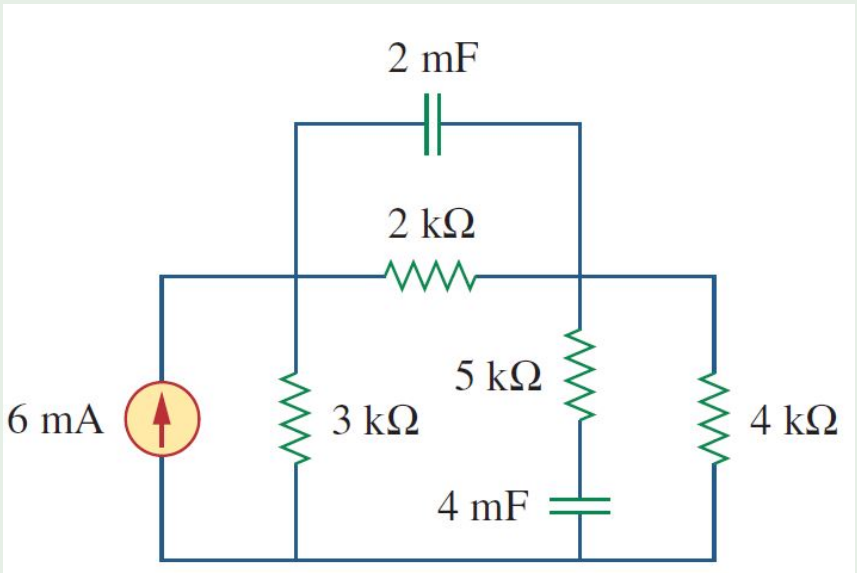
$$i(t) = \begin{cases} 10mA & 0 < t < 1 \\ -10mA & 1 \leq t < 3 \\ 10mA & 3 \leq t < 4 \\ 0 & \text{otherwise} \end{cases} \tag{2}$$



Examples

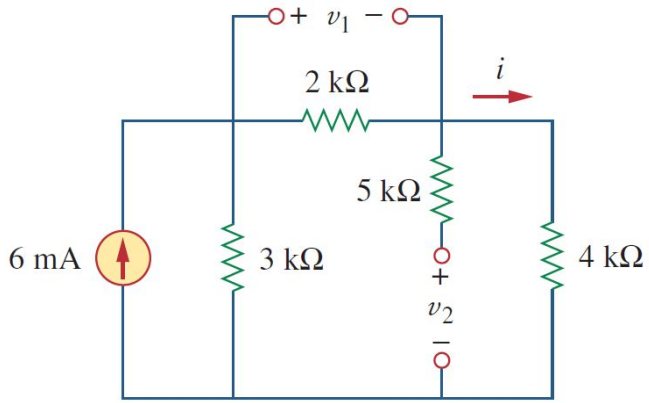
Example

Obtain the energy stored in each capacitor in the figure below under DC conditions.



Examples

Solution: Under DC conditions, we replace each capacitor with an open circuit:



The current through the series combination of the $2k\Omega$ and $4k\Omega$ resistors is obtained as $i = \frac{3}{3+2+4}6mA = 2mA$. Therefore, the voltages v_1 and v_2 across the capacitors are $v_1 = 2000i = 4V$ and $v_2 = 4000i = 8V$ and the energies stored in them are

$$w_1 = \frac{1}{2}C_1 v_1^2 = \frac{1}{2}(2 \times 10^{-3})4^2 = 16mJ$$

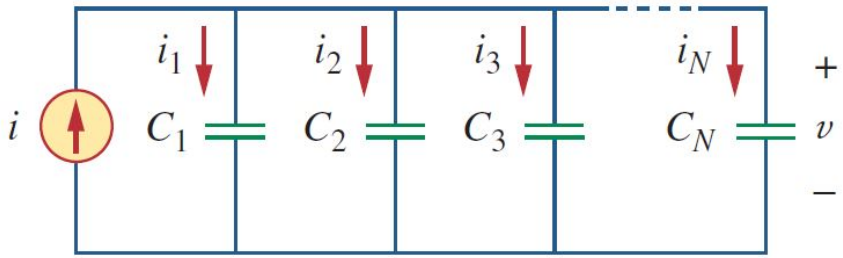
$$w_2 = \frac{1}{2}C_2 v_2^2 = \frac{1}{2}(4 \times 10^{-3})8^2 = 128mJ$$

Series and Parallel Capacitors

- We know from resistive circuits that the series-parallel combination is a powerful tool for reducing circuits.
- This technique can be extended to series-parallel connections of capacitors, which are sometimes encountered.
- We desire to replace these capacitors by a single equivalent capacitor C_{eq}.

Series and Parallel Capacitors

- In order to obtain the equivalent capacitor of N capacitors in parallel, consider the circuit in the below figure. Note that the capacitors have the same voltage across them.



Applying KCL, we get

$$i = i_1 + i_2 + \ldots + i_N$$

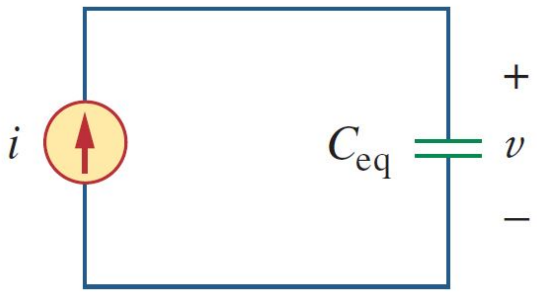
Furthermore, for each capacitor we have $i_k = C_k dv/dt$ $(k = 1, \ldots, N)$.

Series and Parallel Capacitors

Therefore,

$$i = C_1 \frac{dv}{dt} + C_2 \frac{dv}{dt} + \ldots + C_N \frac{dv}{dt}$$
$$= \left(\sum_{k=1}^{N} C_k \right) \frac{dv}{dt} = C_{eq} \frac{dv}{dt}$$

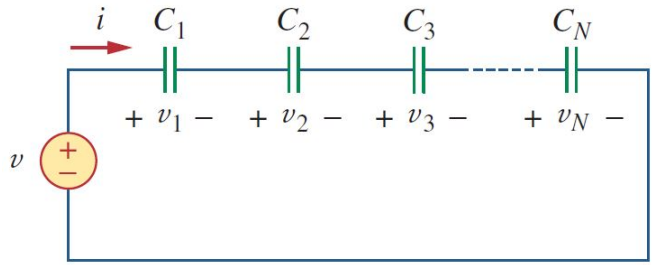
where $C_{eq} = C_1 + C_2 + \ldots + C_N$. **The equivalent capacitance of N parallel-connected capacitors is the sum of the individual capacitances.**



We observe that capacitors in parallel combine in the same manner as resistors in series.

Series and Parallel Capacitors

- We now obtain equivalent capacitance of N capacitors. To this end, consider the circuit illustrated in the figure below.



- Note that the same current i flows (and consequently the same charge) through the capacitors. Applying KVL, we have

$$v = v_1 + v_2 + \ldots + v_N$$

- However, for each capacitor we have $v_k = \frac{1}{C_k}\int_{t_0}^{t} i(\tau)d\tau + v_k(t_0)$. Therefore

Series and Parallel Capacitors

$$
\begin{aligned}
v &= \frac{1}{C_1}\int_{t_0}^{t} i(\tau)d\tau + v_1(t_0) + \ldots + \frac{1}{C_N}\int_{t_0}^{t} i(\tau)d\tau + v_N(t_0) \\
&= \left(\frac{1}{C_1} + \ldots + \frac{1}{C_N}\right)\int_{t_0}^{t} i(\tau)d\tau + v_1(t_0) + v_2(t_0) + \ldots + v_N(t_0) \\
&= \frac{1}{C_{eq}}\int_{t_0}^{t} i(\tau)d\tau + v(t_0)
\end{aligned}
$$

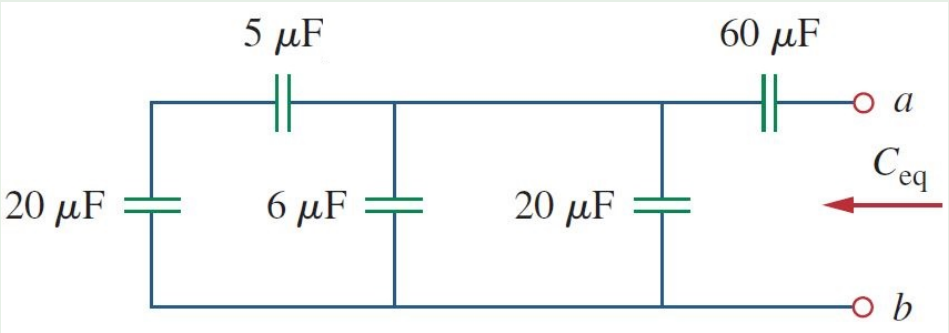
where $\frac{1}{C_{eq}} = \frac{1}{C_1} + \ldots + \frac{1}{C_N}$. Note that the initial voltage $v(t_0)$ across C_{eq} is required by KVL to be the sum of the capacitor voltages at t_0, i.e., $v(t_0) = v_1(t_0) + \ldots + v_N(t_0)$

- **The equivalent capacitance of series-connected capacitors is the reciprocal of the sum of the reciprocals of the individual capacitances.**
- Note that capacitors in series combine in the same manner as resistors in parallel.

Example

Example

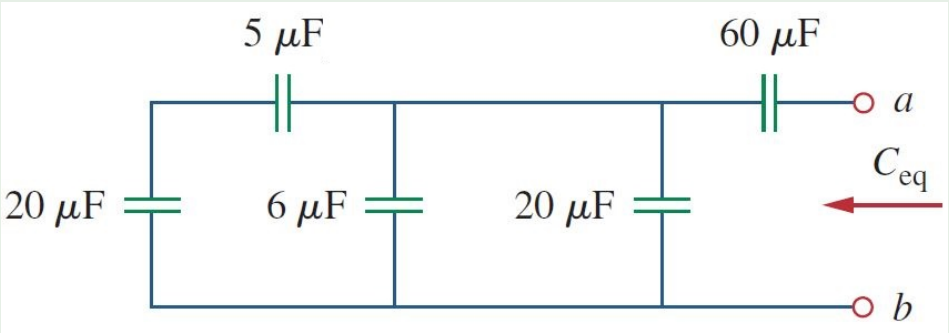
Find the equivalent capacitance seen between terminals a and b of the circuit in the below figure.



Example

Example

Find the equivalent capacitance seen between terminals a and b of the circuit in the below figure.



Answer: $20\mu F$.